ATM Simulator

Program Information

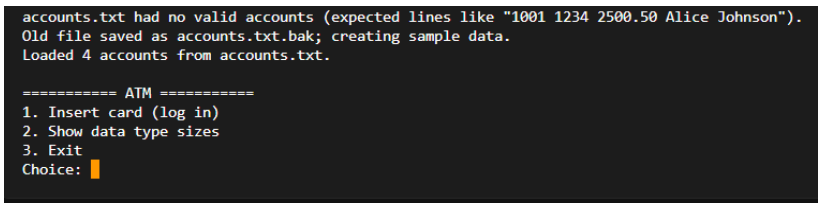
- This program is an ATM simulator that features a main menu with user choice input
    - Displays options for account log in and shows data types sizes
- When program runs for the first time, accounts.txt file will be created with sample data that includes account information
    - It stores data such as accounts with PINs, and transaction history, new accounts can be created by editing the accounts.txt file (Max: 100)
    - The user will be allowed three attempts when entering account PIN
    - transactions.txt file will be created once user has logged in to an account
- After logging in, you can check balance, deposit and withdraw (Limit: $500), or transfer funds to other existing accounts
    - Transaction history is logged during current session and refreshes after new session

Concepts Demonstrated

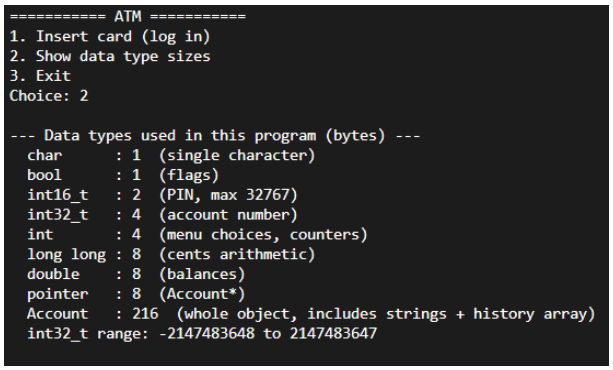
- Data Types:
    - int16_t/int32_t
    - double/char/bool
    - sizeof()
- File Types:
    - accounts.txt (data)
    - transactions.txt (log)
- Pointers:
    - Account* returned by search, pointer-based transfer
- Arrays:
    - Fixed-size Account array, per-account history array
- Binary Search:
    - Bank::findAccount (array sorted by account number)
- Strings:
    - Std::string names, parsing
- File I/O:
    - ifstream/ofstream (load, save, log)
- Classes:
    - Account, Bank, ATM

Output Screenshots

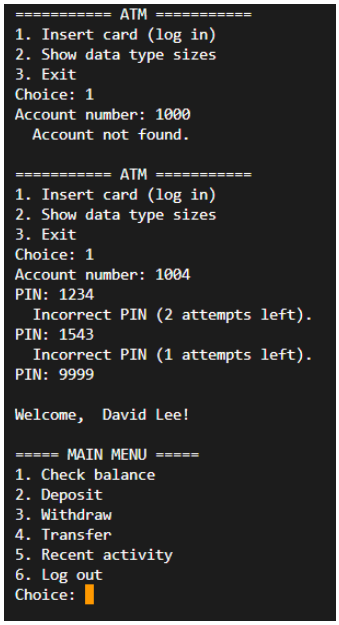
First Program Run:



Data Types Display Option:



Existing Account Log In:



Newly Created Account Log In:

```
========== ATM ==========
1. Insert card (log in)
2. Show data type sizes
3. Exit
Choice: 1
Account number: 1005
PIN: 9341

Welcome,  Carl Johnson!

===== MAIN MENU =====
1. Check balance
2. Deposit
3. Withdraw
4. Transfer
5. Recent activity
6. Log out
Choice:
```

Account Balance, Deposit, Withdrawal, and Recent Activity:

```
 4. Transfer
 5. Recent activity
 6. Log out
 Choice: 1
   Balance: $5353351.53

 ===== MAIN MENU =====
 1. Check balance
 2. Deposit
 3. Withdraw
 4. Transfer
 5. Recent activity
 6. Log out
 Choice: 2
 Deposit amount: $1000
   Deposited $1000.00. New balance: $5354351.53

 ===== MAIN MENU =====
 1. Check balance
 2. Deposit
 3. Withdraw
 4. Transfer
 5. Recent activity
 6. Log out
 Choice: 3
 Withdraw amount: $550
   Limit per transaction is $500.00.

 ===== MAIN MENU =====
 1. Check balance
 2. Deposit
 3. Withdraw
 4. Transfer
 5. Recent activity
 6. Log out
 Choice: 5

 --- Recent activity (this session) ---
   Deposit: +$1000.00
```

Transfer Funds to Existing Account:

```
2. Deposit
3. Withdraw
4. Transfer
5. Recent activity
6. Log out
Choice: 4
Transfer to account number: 1004
Transfer amount: $100000
  Transferred $100000.00 to   David Lee. New balance: $5252351.53

===== MAIN MENU =====
1. Check balance
2. Deposit
3. Withdraw
4. Transfer
5. Recent activity
6. Log out
Choice: 6
Goodbye,   Carl Johnson. Please take your card.

========== ATM ==========
1. Insert card (log in)
2. Show data type sizes
3. Exit
Choice: 1
Account number: 1004
PIN: 9999

Welcome,   David Lee!

===== MAIN MENU =====
1. Check balance
2. Deposit
3. Withdraw
4. Transfer
5. Recent activity
6. Log out
Choice: 1
  Balance: $100000.00
```